

Complex Analysis

5.4 Complex Variable (Functions)

$$Z = x + iy, \quad x, y \in R \quad f(z) = u(x, y) + iv(x, y), \quad u, v \in R$$

Analytic function: A function that is defined and differentiable in a domain D is known as the Analytic function.

Example: $f(z) = z, \sin z$

Non-Analytic function: If a function is not continuous or differentiable then it is not analytic.

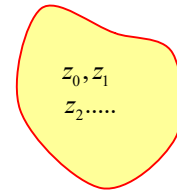
Example: $f(z) = \bar{z}$

Necessary conditions:

Cauchy Riemann formula:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} : \text{Cartesian form}$$

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r} : \text{Polar form}$$



Example: Solve the following functions and tell whether the function is analytic or non-analytic.

(a) $f(z) = z$

Solution: $f(z) = z = x + iy = u + iv$, $u = x, v = y$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 1, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 0$$

∴ The function follows the Cauchy Riemann condition, So the function is analytic.

(b) $f(z) = z^2$

Solution: $f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$, $u = x^2 - y^2, v = 2xy$

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial v}{\partial y} = 2x, \quad \frac{\partial u}{\partial y} = -2y, -\frac{\partial v}{\partial x} = -2y \Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \& \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

∴ The function follows the Cauchy Riemann condition, So the function is analytic.

(c) $f(z) = z^3$

Solution: $f(z) = (x + iy)^3 = x^3 + i^3 y^3 + 3x^2 iy + 3xi^2 y^2 = x^3 - iy^3 + 3x^2 iy - 3xy^2$

$$u = (x^3 - 3xy^2), \quad v = (3x^2 y - y^3)$$

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial v}{\partial y} = 3x^2 - 3y^2; \quad \frac{\partial u}{\partial y} = -6xy, \quad -\frac{\partial v}{\partial x} = -6xy$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \& \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \text{ So the function is analytic.}$$

In general: $f(z) = a_0 + a_1 z + a_2 z^2 + \dots$ which is **Taylor series** is an **Analytic function** because

$z, z^2, z^3, \dots$ all are analytic functions.

$$\text{Taylor series: } f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

(d) $f(z) = \sin z$

Solution: $f(z) = \sin(x + iy) = \sin x \cos iy + \cos x \sin iy = \sin x \cosh y + i \cos x \sinh y$

$$u = \sin x \cosh y, \quad v = \cos x \sinh y$$

$$\frac{\partial u}{\partial x} = \cos x \cosh y, \quad \frac{\partial v}{\partial y} = \cos x \cosh y, \quad \frac{\partial u}{\partial y} = \sin x \sinh y, \quad -\frac{\partial v}{\partial x} = \sin x \sinh y$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ \& \; } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \text{ So the function is analytic.}$$

(e) $f(z) = \bar{z}$

Solution: $f(z) = \bar{z} = x - iy, \quad u = x, v = -y$

$$\frac{\partial u}{\partial x} = 1, \frac{\partial v}{\partial y} = -1, \frac{\partial u}{\partial y} = 0, \frac{\partial v}{\partial x} = 0$$

$$\Rightarrow \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y} \text{ \& \; } \frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x}, \text{ So the function is not analytic.}$$

The special thing about $\bar{z}$:

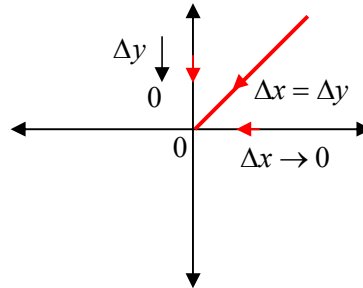
$$f(z) = \bar{z}, \quad f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\overline{z + \Delta z} - \bar{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\bar{z} + \overline{\Delta z} - \bar{z}}{\Delta z}$$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = -1, \quad \lim_{\Delta y \rightarrow 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = 1$$

$$\lim_{\Delta y \rightarrow \Delta x} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = \frac{1-i}{1+i}$$



$\therefore$ We get different values. So If there is a complex variable $\bar{z}$ in the function then the function will be non-analytic.

Example: $f(z) = (-x + iy + 7)^7$

Solution: $f(z) = -(x - iy - 7)^7 = -(\bar{z} - 7)^7$

The function will be non-Analytic.

Example: $f(z) = z\bar{z}$

Solution: $f(z) = z\bar{z} = (x + iy)(x - iy) = (x^2 + y^2) + 0i$

$$u = x^2 + y^2, \quad v = 0$$

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial v}{\partial y} = 0, \frac{\partial u}{\partial y} = 2y, \frac{\partial v}{\partial x} = 0$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x} \text{ So the function is non-analytic.}$$

Example: $f(z) = e^z$ This is an analytic function.

Solution: $f(z) = e^z = e^{(x+iy)} = e^x \cdot e^{iy} = e^x \cdot [\cos y + i \sin y] = e^x \cos y + i e^x \sin y$

$$u = e^x \cos y, \quad v = e^x \sin y$$

$$\frac{\partial u}{\partial x} = e^x \cos y, \frac{\partial v}{\partial y} = e^x \cos y, \quad \frac{\partial u}{\partial y} = -e^x \sin y, \frac{\partial v}{\partial x} = e^x \sin y$$

So this is an analytic function.

Example: $f(z) = \ln z$

Solution: $f(z) = \ln z, \quad z = r e^{i\theta} \text{ so } f(z) = \ln r e^{i\theta}$

$$f(z) = \ln r + i \theta = \ln r + i \theta, \quad u = \ln r, v = \theta$$

$$\frac{\partial u}{\partial r} = \frac{1}{r}, \frac{\partial v}{\partial \theta} = 1 \Rightarrow \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \& \quad \frac{\partial u}{\partial \theta} = 0, \frac{\partial v}{\partial r} = 0 \Rightarrow \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$$

So this is an analytic function.

Calculation of u or v if either of them is known:

Example: If the real part (u) of the following function is given then find the imaginary part (v).

(a) $u = x^2 - y^2$

Solution: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 2x \Rightarrow \partial v = 2x \partial y \Rightarrow v = 2xy + g(x) \quad (1)$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -2y \Rightarrow \partial v = 2y \partial x \Rightarrow v = 2yx + h(y) \quad (2)$$

From equations (1) and (2), the imaginary part will be $v = 2xy$.

(b) $u = x^2 - y^2 - y$

Solution: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 2x \Rightarrow \partial v = 2x \partial y \Rightarrow v = 2xy + g(x) \quad (1)$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -2y - 1 \Rightarrow \frac{\partial v}{\partial x} = 2y + 1 \Rightarrow v = \int (2y + 1) \partial x \quad (2)$$

$$v = 2yx + x + h(y)$$

From equations (1) and (2), the imaginary part will be $v = 2xy + x$.

(c) $u = x + \frac{1}{2}(x^2 - y^2)$

Solution: $u = x + \frac{1}{2}(x^2 - y^2)$

$$\frac{\partial u}{\partial x} = 1 + x = \frac{\partial v}{\partial y}, \quad \partial v = (1+x)\partial y, \quad v = y + xy + g(x)$$

$$\frac{\partial u}{\partial y} = -y = -\frac{\partial v}{\partial x} \Rightarrow \frac{\partial v}{\partial x} = y \Rightarrow v = yx + h(y)$$

So $v = xy + y$

(d) $u = \log|z|$

Solution: $u = \log|z|$, $z = re^{i\theta}$, $|z| = |r(\cos\theta + i\sin\theta)|$

$$|z| = \sqrt{r^2(\cos^2\theta + \sin^2\theta)} = \sqrt{r^2} = r, \quad u = \log r$$

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \Rightarrow \frac{1}{r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \Rightarrow v = \theta + g(r), \quad \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r} \Rightarrow 0 = \frac{\partial v}{\partial r} \Rightarrow v = c$$

So $v = \theta$.

(e) $u = x^3 - 3xy^2$

Solution: $u = x^3 - 3xy^2$, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \Rightarrow 3x^2 - 3y^2 = \frac{\partial v}{\partial y}$,

$$v = 3x^2y - \frac{3y^3}{3} = 3x^2y - y^3 + g(x) \quad (1)$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \Rightarrow -6xy = -\frac{\partial v}{\partial x}, \quad v = \frac{6x^2y}{2} = 3x^2y + h(y) \quad (2)$$

From equation (1) and (2), $v = 3x^2y - y^3$.

(f) $u = e^{-x} \sin 2y$

Solution: $u = e^{-x} \sin 2y$, $\frac{\partial u}{\partial x} = -e^{-x} \sin 2y = \frac{\partial v}{\partial y}$

$$\int \partial v = -e^{-x} \int \sin 2y \partial y, \quad v = e^{-x} \frac{\cos 2y}{2} + g(x)$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \Rightarrow e^{-x} 2 \cos 2y = -\frac{\partial v}{\partial x}, \quad \int \partial v = -\cos 2y \int e^{-x} \partial x$$

$$v = 2 \cos 2y e^{-x} + h(y)$$

So $v = 2 \cos 2y e^{-x} + e^{-x} \frac{\cos 2y}{2}$.

(g) $u = \cos x \cosh y$

Solution: $u = \cos x \cosh y$, $\frac{\partial u}{\partial x} = -\sin x \cosh y = \frac{\partial v}{\partial y}$, $v = -\sin x \sinh y + g(x)$ (1)

$\frac{\partial u}{\partial y} = \cos x \sinh y = -\frac{\partial v}{\partial x}$, $v = -\sin x \sinh y + h(y)$ (2)

So $v = -\sin x \sinh y$

(h) $u = \log \sqrt{x^2 + y^2}$

Solution: $u = \log r$, $\left[r = \sqrt{x^2 + y^2} \right]$, $\frac{\partial u}{\partial r} = \frac{1}{r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \Rightarrow v = \theta + g(r)$

$\frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r} \Rightarrow v = c$, So $v = \theta$, $\theta = \tan^{-1}\left(\frac{y}{x}\right)$, $v = \tan^{-1}\left(\frac{y}{x}\right)$

(i) $u = 2x(1-y)$

Solution: $u = 2x(1-y)$, $\frac{\partial u}{\partial x} = (2-2y) = \frac{\partial v}{\partial y}$, $v = 2y - y^2 + g(x)$ (1)

$\frac{\partial u}{\partial y} = -2x = -\frac{\partial v}{\partial x} \Rightarrow v = \frac{2x^2}{2} + h(y)$ (2)

$v = x^2 - y^2 + 2y$

(j) $u = \left(r - \frac{1}{r}\right) \sin \theta$

Solution: $u = \left(r - \frac{1}{r}\right) \sin \theta$, $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} = \sin \theta + \frac{\sin \theta}{r^2}$, $r \sin \theta + \frac{\sin \theta}{r} = \frac{\partial v}{\partial \theta}$

$\frac{\partial v}{\partial \theta} = r \sin \theta + \frac{\sin \theta}{r}$

$\int \partial v = \int r \sin \theta \partial \theta + \int \frac{\sin \theta}{r} \partial \theta$, $v = -r \cos \theta - \frac{\cos \theta}{r} + g(r)$ (1)

$\frac{\partial u}{\partial \theta} = r \cos \theta - \frac{\cos \theta}{r} = -r \frac{\partial v}{\partial r}$, $\frac{\cos \theta}{r} - r \cos \theta = r \frac{\partial v}{\partial r}$, $\frac{\cos \theta}{r^2} - \cos \theta = \frac{\partial v}{\partial r}$

$v = \int \frac{\cos \theta}{r^2} \partial r - \int \cos \theta \partial r$, $v = -\frac{\cos \theta}{r} - r \cos \theta + h(\theta)$ (2)

So $v = -\frac{\cos \theta}{r} - r \cos \theta$